

SANJEEV RATNANI

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EDUCATION

Sardar Patel Institute of Technology (SPIT), Mumbai

B.Tech. in Computer Engineering

Nov. 2022 - June 2026

CGPA: 9.51/10.0

Relevant Coursework:

System Design, Distributed Computing, Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, OOP, Artificial Intelligence & Machine Learning, Cloud Computing, Generative AI, Building LLMs (GPTs)

S.P. Jain Institute of Management and Research (SPJIMR)

Minors in Finance and Management

Jan. 2024 - Dec. 2025

PROFESSIONAL EXPERIENCE

Technology Intern [Application Developer]

Russell Investments - A top global asset management firm with \$330B+ AUM

May 2025 - July 2025

Mumbai, Maharashtra

- Architected **AI-powered enterprise API testing platform** that autonomously generates 1000s of **schema-compliant test cases** from OpenAPI specs, utilizing **distributed computing architecture** with multithreading, multiprocessing, & async I/O patterns
- Engineered **intelligent fuzzing engine** with real-time parameter validation, adaptive header injection, and **GenAI-driven coverage optimization**, achieving **10x performance improvement** over legacy testing frameworks
- Delivered **enterprise-scale API validator** with **fault-tolerant design** and robust error handling

PROJECT LEADERSHIP & PRODUCT DEVELOPMENT

Lead Product Developer & System Architect

AI-Powered Student Focus Monitoring Platform

Feb. 2025 - Present

Independent Project

- Built **production-grade AI application** with ML-based computer vision for attention tracking, resulting in **37% measurable improvement** in study efficiency across beta user base
- Designed and deployed a **scalable product ecosystem** with Flutter, Firebase, real-time analytics, and parent monitoring, scaling to **250+ users** with **93% satisfaction** via performance tuning and data-driven UX iterations

Full-Stack Platform Developer

Enterprise Learning Management System - NKCCA

Mar 2025 - Present

Contract Development

- Architected **multi-platform learning ecosystem** with Flutter tablet application and React.js admin portal, designed for **enterprise-scale deployment**
- Built a **distributed backend** with Firebase supporting **5000+ concurrent users** at sub-second latency, and integrated **AWS IVS** for low-latency adaptive educational streaming via global CDN

Software Developer & UX Engineer

Mental Health Platform - Sattvastha.AI

Apr. 2025 - Present

Product Development

- Developing **intelligent psychological assessment platform** with adaptive questionnaire algorithms
- Implementing **secure healthcare-compliant architecture** using Next for web & React Native + Expo for app with Firebase backend, designed for handling sensitive mental health data with encryption and audit trails

Research Intern & App Developer

Medical AI Research - SPIT-KEM Hospital Collaboration

Jan. 2025 - Present

Research Project

- Contributing to **medical AI research initiative** for early-stage diabetic foot detection using CV and DL classification models
- Building **evidence-based rehabilitation platform** with adaptive exercise tracking, progress analytics, & clinical data integration

RECOGNITION & INNOVATION

Problem-Solving Excellence | 10x National & International Hackathon Winner Including

- Smart India Hackathon 2024 - National Winner** | Selected as 1 of 315 winning teams from **500,000+ participants nationwide**

Patent Inventor | "PETHUB - Integrated Pet Care System" - Published by Office of Controller General of Patents, India

(Application No. 202421102429, Published 31/01/2025) - Recognized for innovative system design and technical contribution

TECHNICAL PROJECTS & SOLUTIONS

Healthwise - Remote Patient Monitoring System | Flutter, Firebase, TensorFlow, IoT Integration

Feb. 2025

- Engineered **comprehensive healthcare platform** with real-time vitals, AI health insights, and cross-platform deployment
- Built **3-tier application suite**: Patient mobile app, Doctor dashboard with AI-assisted diagnosis, and WatchOS companion app with seamless wearable integration
- Implemented **secure healthcare data pipeline** with HIPAA-compliant architecture and real-time analytics engine

Dhara - Industrial Safety Platform | Flutter, React.js, Node.js, Python, IoT, Offline-First Architecture

Dec. 2024

- Developed **mission-critical safety system** for coal mining operations, improving operational efficiency by **50%** through real-time monitoring and automated alerts
- Architected **offline-first mobile solution** with Hive local database, ensuring **80% uptime improvement** in low-connectivity
- Integrated **IoT sensor network** with automated alert systems, multi-language support, and real-time data synchronization

TECHNICAL EXPERTISE

Programming & Scripting: C/C++, Java, Dart, Python, JavaScript/TypeScript

System Design & Architecture: Distributed Systems, Microservices, System Design Patterns, Scalability Design

Full-Stack Development: React.js, Node.js, Express.js, RESTful APIs, GraphQL, Flutter, React Native

AI & Machine Learning: TensorFlow, Computer Vision, Neural Networks, GenAI Integration, AI-Augmented Development

Database & Storage: Firebase, MongoDB, MySQL, Redis, Cloud Storage Solutions, Data Pipeline Design

Cloud & DevOps: AWS, Firebase, Docker, CI/CD, Performance Optimization, Auto-scaling

Mobile & Cross-Platform: Flutter, React Native, Offline-First Architecture, Performance Optimization